

XIYUAN (TOMMY) GUO

B.A. Candidate in Computer Science, University of Virginia
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RESEARCH FOCUS

Convex optimization (ADMM); projection-based imputation algorithms; high-performance C++ solver engineering; computational imaging; quantum software infrastructure.

EDUCATION

University of Virginia
B.A. in Computer Science

Expected May 2027
Charlottesville, VA

- **GPA:** 3.89/4.00
- **Advanced Physics & Quantum Coursework:** Graduate Quantum Computing (PHYS 5880, in progress); Quantum Information & Computation (completed Caltech PH219A curriculum); Theoretical Quantum Mechanics (completed Griffiths and Sakurai curricula).

Kimball Union Academy
High School Diploma

2024
Meriden, NH

GRANTS AND FELLOWSHIPS

- **Double Hoo Research Grant**, University of Virginia 2026
Secured competitive grant funding to drive independent, interdisciplinary computational research alongside graduate mentorship.

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

- Guo, X., [Co-authors], Ke, L. "Transition-metal magnetic anisotropy and its relationship to structural motifs." *[Manuscript in preparation]*.
- Guo, X., [Co-authors], Liu, M. "Joint Spectro-Polarimetric Lensless Reconstruction via Conic-Constrained ADMM." *[Manuscript in preparation]*. Code available on request.
- Guo, X., [Co-authors], Tong, X.C. "PRISM: Lagrangian Projection Engines for Multivariate Longitudinal Missing Data Imputation." *[Manuscript in preparation]*. Code: github.com/xguot/prism.

PROFESSIONAL RESEARCH EXPERIENCE

Research Assistant

Yale University (Prof. Mengxia Liu's Group)

Feb 2026 – Present

New Haven, CT

- Engineered a joint spectro-polarimetric forward model to recover a 324-channel scene datacube (spatial $\times$ 81 wavelengths $\times$ 4 Stokes channels) from a single multiplexed lensless speckle image.
- Formulated a conic-constrained inverse problem mathematically isomorphic to quantum state tomography (QST): the Poincaré-Bloch map $S \mapsto \rho = \frac{1}{2}(I + S \cdot \sigma)$ sends each Stokes vector to a single-qubit density matrix, and the Lorentz cone constraint ($S_0 \geq ||S_{1:3}||$) enforced in the reconstruction is precisely the density-matrix positivity condition $\rho \succeq 0$.
- Architected a GPU-accelerated consensus ADMM solver in PyTorch, executing exact joint proximal operators for L_1 spectral sparsity and closed-form Sherman-Morrison matrix inversions, scaling massive tensor operations across SLURM HPC clusters.
- Calibrated optical point-spread-function (PSF) libraries utilizing spatial Gram-Schmidt orthogonalization to eliminate channel cross-talk and non-negative least squares (NNLS) for spectral unmixing.

Research Assistant

UVA (Prof. Liqin Ke's Group)

Oct 2025 – Present

Charlottesville, VA

- Applied Density Functional Theory (DFT) to investigate transition-metal magnetic anisotropy and its fundamental relationships to structural motifs within complex crystal lattices.
- Created immersive, browser-based data visualizations using Three.js and JSmol, generating interactive 3D atomic models to facilitate real-time structural inspection and theoretical validation.
- Engineered a cloud-native machine learning pipeline utilizing Go and Next.js, enabling global academic collaborators to securely share specialized datasets and execute predictive models on quantum material properties.
- Designing RareEarth DB, a rare-earth-focused analogue of the Materials Project: a MongoDB document architecture linking 4,648 curated magnetic-anisotropy measurements to ICSD/Pearson crystal structures, with a schema designed for Fireworks-orchestrated DFT workflows on UVA HPC; DOE funding proposal in preparation.

Research Assistant

UVA BAL Lab (Prof. Xin Cynthia Tong's Group)

Apr 2026 – Present

Charlottesville, VA

- Engineered PRISM, a constrained optimization framework for multivariate longitudinal missing data imputation: an initial imputation is projected onto the FIML model-implied (mean, covariance) structure under a regularized objective that counteracts the variance attenuation inherent in unconstrained conditional-expectation imputation.
- Built the solver engine in C++17 (RcppArmadillo): projected gradient descent with Armijo backtracking line search, exact mean-constraint enforcement via per-column zero-sum gradient projection, and KKT convergence certification computing per-column Lagrange multiplier estimates at every iterate.

Research Intern

UVA Biocomplexity Institute

May 2026 – Aug 2026

Charlottesville, VA

- Formulated massive-scale network simulations to dynamically model the stochastic spread of viral pathogens, implementing SEIR compartmental cascade models to capture heterogeneous transmission dynamics.
- Engineered Graph Neural Network (GNN) architectures utilizing spatiotemporal message-passing frameworks to extract predictive topological features from dynamic contact networks.

OPEN SOURCE SOFTWARE

- **stim (Contributor):** Authored pull request (#1099) for the open-source quantum circuit simulation library, contributing to high-performance stabilizer circuit rendering and execution frameworks.
- **smriti / PRISM:** Software framework for automated longitudinal missing data imputation. Executes an architecture combining initialization with a custom C++ Lagrangian projection engine to strictly preserve the structural variance of the target covariance manifold.
- **sanjaya:** Automated academic literature extraction pipeline utilizing the OpenAlex Academic Graph API, Scrapy, and Playwright to extract bilingual data, exporting structured datasets for downstream interdisciplinary research.
- **difi:** High-performance Go-based Git diff terminal user interface utilizing the Bubble Tea framework; achieved official Homebrew Core inclusion, 342+ GitHub stars, and a #5 ranking on Hacker News.

HONORS AND AWARDS

- **Double Hoo Research Grant**, University of Virginia, 2026
- **Dean's List of Distinguished Students**, University of Virginia
- **Silver Division**, USA Computing Olympiad (USACO)

TECHNICAL SKILLS

- **Math & Physics:** Advanced Theoretical Quantum Mechanics, Quantum Information Theory, Convex Optimization, Matrix Calculus, Linear Algebra.
- **Machine Learning & Computing:** PyTorch/CUDA, JAX, Scikit-Learn, Pandas, RcppArmadillo.
- **Languages:** Python, C/C++, Julia, Rust, Go, TypeScript, R, SQL, Bash, $\text{\LaTeX}$.
- **Infrastructure & Cloud:** Linux (openSUSE/RHEL), Docker, K3s Kubernetes, Slurm HPC, Git.

COMMUNITY SERVICE

Volunteer Firefighter

Plainfield Fire Department

2022 – 2024

Plainfield

- Responded to over 200 emergency medical, fire, and rescue calls annually, operating effectively in high-stress environments to provide critical incident mitigation and community support.